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🎓 EDUCATION

- M.S. in Data Science** 05/2025 | Washington, DC
George Washington University
Coursework: Machine Learning, Algorithm Design, Data Mining, Quantitative Risk Management
- M.S. in Management** 03/2021 | Tokyo, Japan
The University of Tokyo
Coursework: International Management, Business Economics, Corporate Strategy
- B.S. in Management & Information Sciences** 08/2017 | Chiba, Japan
Josai International University
- B.S. in Marketing & Japanese** 08/2015 | Dalian, China
Dalian University of Foreign Languages

💼 PROFESSIONAL EXPERIENCE

- Machine Learning Engineer** 07/2025 – Present | Remote
MyEdMaster, LLC
- Developed core reasoning methodology of Large Language Model (LLM) learning assistant, enabling AI to better understand and evaluate student thinking through adaptive feedback and self assessment-driven interactions in Natural language processing (NLP) setting.
 - Experimented with algorithmic strategies including Retrieval Augmented Generation (RAG), Chain of Thought (CoT) prompting, and function calling, leveraging OpenAI, Gemini, and Mistral models through API integrations; analyzed model behaviors, refined evaluation logic, and improved accuracy through structured benchmarking and prompt optimization.
 - Integrated system components across front end, model service, and database layers to establish unified data flow, experiment tracking, and rapid iteration for ongoing model improvement.
 - Extended and validated framework beyond original domain, demonstrating scalability and achieving over 90% accuracy on internal benchmarks, significantly outperforming baseline LLMs.
 - Partnered with academic institutions to validate and promote framework, co-authoring paper under review, supporting conferences, and contributing to business development efforts.
- Instructional Assistant, Data Visualization & Data Warehousing** 08/2024 – 05/2025 | Washington, DC
George Washington University
- Led hands-on labs and guided graduate students from non-technical backgrounds to master SQL, MongoDB, and Tableau, from basic querying to building interactive dashboards for large data sets.
 - Taught data modeling and integration for analytics, including star and snowflake schema design, data warehousing concepts, and query optimization; coached teams to translate requirements into clear visual narratives and metric definitions.
 - Partnered with professor to refine course structure and lab materials, improving ETL workflows and connecting technical training with analytical communication and business decision making.

TECHNICAL SKILLS & LANGUAGES

Languages — Python, SQL, R, JavaScript (React, Node.js), TypeScript

AI / ML — LangChain, OpenAI & Gemini APIs, scikit-learn, TensorFlow, XGBoost

Data & Infrastructure — PostgreSQL, MySQL, MongoDB, Airflow, Docker, AWS

Dev Tools — Flask/FastAPI, Git, CI/CD, RAG, Prompt Engineering

Visualization — Tableau, Power BI, Plotly Dash, Excel

PROJECT EXPERIENCE

EdTech Support Chatbot Development 03/2025 – 06/2025

- Designed and implemented intelligent support chatbot for educational platforms, evolving from rule-based Python + NLTK system to LLM-driven model with ChatGPT API integration.
- Applied NLP techniques and prompt engineering to improve intent recognition, dialogue flow, and context retention, enhancing user experience in multi-turn conversations.
- Collaborated on product documentation and UX design, drafting PRDs, creating wireframes, and iterating through A/B testing and feedback analysis to refine chatbot functionality.

Capstone: Mental Health Demand & Supply Analysis & Forecasting 01/2025 – 05/2025

- Tackled mismatch between mental health service demand and provider supply across U.S. states by integrating Google Trends (400+ keywords, 2000-2024) and HRSA data to surface real-time unmet needs.
- Designed time series forecasting pipeline with 100+ keyword-derived features; built and tuned XGBoost models to predict anxiety and depression trends, applying SHAP explainability to highlight top predictors.
- Delivered highly interactive, state-level web dashboard  using Plotly Dash for public deployment, enabling dynamic exploration of search trends, regional comparisons, and predictor-driven insights.

Wallet Wizard: Personal Finance Tracker 01/2025 – 03/2025

- Designed and developed full-stack personal budgeting app to help users track expenses, set spending goals, and gain real-time insights into their financial health.
- Built RESTful APIs and integrated React, Next.js frontend with a Node.js and PostgreSQL backend, enabling real-time expense tracking and seamless front-to-back data flow and live budget updates.
- Led agile development workflows, managing GitHub version control, pull requests, and issue tracking, and delivered functional MVP within eight weeks, demonstrating key features during live demo.

Machine Learning for Optimizing Strategic NIH Funding Allocation 01/2024 – 05/2024

- Tackled challenge of inefficient NIH grant classification by developing machine learning models to improve how funding requests are categorized, potentially enabled NIH to make smarter investment decisions in critical health research areas.
- Analyzed 118,000+ NIH grant records from FY18–FY23 to uncover patterns in funding distribution, focusing on key career pipeline grants essential for researcher development.
- Developed optimized classification models (Random Forest, Logistic Regression) using Scikit-learn, achieving 92.96% accuracy to categorize NIH grant types.
- Produced visual insights with Tableau, identifying underfunded research domains and emerging fields.

Automated Data Pipeline for Enterprise Data Security 06/2024 – 08/2024

- Built flexible data management framework to address challenge of securely transforming and transferring sensitive enterprise data across relational (MySQL) and non-relational (MongoDB) systems, minimizing risks in lower-security environments.
- Designed and implemented modular Python packages for data sanitization, obfuscation, minimization, and restoration, integrating them into automated workflows using Apache Airflow DAGs.